

1 System Model and Problem Formulation

1.1 System Scenario

We consider a 5G-Advanced frequency-division duplexing (FDD) massive-MIMO downlink in which a base station (gNB) equipped with M antennas serves a set of users $\mathcal{K} = \{1, \dots, K\}$. The gNB employs a predefined beamforming codebook $\mathcal{B} = \{\mathbf{w}_1, \dots, \mathbf{w}_B\}$, where $\mathbf{w}_b \in \mathbb{C}^M$ denotes the b th beamforming vector. The system operates over a set of resource blocks $\mathcal{N} = \{1, \dots, N\}$. The users receive task-oriented semantic information rather than treating all transmitted packets as equally valuable. Examples include object features for augmented reality, visual information for edge-assisted inference, robot-control messages, vehicular perception updates, and event notifications. A high-priority semantic update may significantly affect the downstream task, whereas the loss of a low-priority update may have little impact. This observation motivates a different treatment of channel-state information (CSI). Conventional systems normally obtain CSI according to channel quality, mobility, or a fixed feedback periodicity. Such an approach does not consider whether improving CSI accuracy actually benefits the semantic task currently waiting for transmission. For example, the gNB may spend substantial feedback resources to obtain an accurate channel estimate even when it is transmitting low-value semantic information. Conversely, an inaccurate predicted beam may severely degrade a critical semantic update. We therefore allow the gNB to jointly determine whether a user should provide fresh CSI, how accurately the user should compress that CSI, whether the gNB can rely on beam prediction instead, which beam should serve the user, which semantic representation should be transmitted, and how the available radio resources should be allocated. The resulting architecture contains four interacting components. First, each user estimates its downlink channel from reference signals using either a conventional or a learned channel estimator. Second, the user extracts a compact task-oriented representation of its CSI instead of always reconstructing the complete channel matrix at the gNB. Third, when fresh CSI feedback is unnecessary, the gNB predicts the user's beam from historical channel and beam observations. Finally, the scheduler jointly allocates feedback, beams, semantic representations, radio resources, and transmit power according to the expected downstream task utility.

1.2 Channel and CSI Estimation Model

Let $\mathbf{h}_k(t) \in \mathbb{C}^M$ denote the downlink channel from the gNB to user k during slot t . The pilot observation at user k is represented as

$$\mathbf{y}_k^p(t) = \mathbf{X}^p \mathbf{h}_k(t) + \mathbf{n}_k^p(t), \quad (1)$$

where $\mathbf{X}^p$ denotes the pilot matrix and $\mathbf{n}_k^p(t)$ denotes the corresponding receiver noise. Each user computes an estimate of its instantaneous channel as

$$\hat{\mathbf{h}}_k(t) = f_\phi\left(\mathbf{y}_k^p(t), \hat{\mathbf{h}}_k(t-1 : t-L_h)\right), \quad (2)$$

where $f_\phi(\cdot)$ represents a channel estimator and L_h denotes the number of historical observations used by the estimator. The proposed framework does not require a particular architecture for $f_\phi(\cdot)$; it may correspond to LS, LMMSE, CNN, GRU, or a lightweight Transformer. Channel estimation therefore acts as an input to the proposed optimization framework rather than constituting the sole contribution of the system.

1.3 Task-Oriented Semantic CSI Representation

Conventional CSI compression attempts to reconstruct $\mathbf{h}_k(t)$ or a corresponding channel matrix as accurately as possible. However, the gNB does not necessarily require every channel coefficient. In a codebook-based beam-management system, the gNB primarily needs enough channel information to identify a high-quality transmission beam. For this reason, we define the instantaneous beam-utility vector of user k as

$$\mathbf{g}_k(t) = [g_{k1}(t), \dots, g_{kB}(t)], \quad (3)$$

where

$$g_{kb}(t) = |\mathbf{h}_k^H(t) \mathbf{w}_b|^2. \quad (4)$$

The semantic CSI encoder aims to preserve the information required to estimate $\mathbf{g}_k(t)$ and rank the candidate beams rather than accurately reconstructing every coefficient of $\mathbf{h}_k(t)$. We define a set of CSI-feedback modes

$$\mathcal{M} = \{0, 1, \dots, M_f\}, \quad (5)$$

where mode $m = 0$ represents *prediction without current CSI feedback*, while $m > 0$ represents progressively richer semantic CSI descriptions. Let

$$x_{km}(t) \in \{0, 1\} \quad (6)$$

indicate whether user k selects feedback mode m at slot t . Each user selects exactly one mode:

$$\sum_{m \in \mathcal{M}} x_{km}(t) = 1, \quad \forall k, t. \quad (7)$$

Let C_m denote the number of feedback bits required by mode m , where

$$C_0 = 0, \quad C_1 < C_2 < \dots < C_{M_f}. \quad (8)$$

For $m > 0$, user k generates a semantic CSI representation

$$\mathbf{z}_k^m(t) = E_\theta^m \left(\hat{\mathbf{h}}_k(t), \hat{\mathbf{h}}_k(t-1:t-L_h) \right), \quad (9)$$

where $E_\theta^m(\cdot)$ denotes a learned encoder associated with feedback resolution m . A progressively refinable implementation may satisfy

$$\mathbf{z}_k^m = [z_{k,1}, \dots, z_{k,d_m}], \quad d_1 < d_2 < \dots < d_{M_f}, \quad (10)$$

so that a higher feedback mode refines rather than replaces the lower-rate representation. The total CSI feedback consumed by all users must satisfy

$$\sum_{k \in \mathcal{K}} \sum_{m \in \mathcal{M}} C_m x_{km}(t) \leq C_{\max}^{\text{FB}}, \quad \forall t, \quad (11)$$

where $C_{\max}^{\text{FB}}$ denotes the feedback capacity available during one scheduling interval.

1.4 Beam Prediction and Semantic CSI Decoding

When user k selects a feedback mode $m > 0$, the gNB estimates the user's beam-utility vector as

$$\hat{\mathbf{g}}_k^m(t) = D_\omega(\mathbf{z}_k^m(t)), \quad (12)$$

where $D_\omega(\cdot)$ denotes the semantic CSI decoder. When $m = 0$, the user does not transmit current CSI. The gNB instead predicts its beam utilities using historical information:

$$\hat{\mathbf{g}}_k^0(t) = P_\psi(\mathbf{z}_k(t-1:t-L_p), b_k(t-1:t-L_p)), \quad (13)$$

where $P_\psi(\cdot)$ denotes the beam predictor and L_p represents its observation window. The gNB then selects a beam according to

$$\hat{b}_k(t) = \arg \max_{b \in \mathcal{B}} \hat{g}_{kb}(t). \quad (14)$$

To measure whether the compressed or predicted CSI preserves the information that matters to beam management, we define the normalized beam-semantic regret as

$$d_k^B(t) = 1 - \frac{\left| \mathbf{h}_k^H(t) \mathbf{w}_{\hat{b}_k(t)} \right|^2}{\max_{b \in \mathcal{B}} \left| \mathbf{h}_k^H(t) \mathbf{w}_b \right|^2 + \epsilon}, \quad (15)$$

where $\epsilon > 0$ avoids numerical instability. A value close to zero indicates that the compressed or predicted CSI retains the information required to identify the best beam. In contrast, a large value indicates a substantial loss of task-relevant channel information. This metric differs from conventional channel NMSE because it evaluates CSI according to its effect on the communication decision.

1.5 Semantic Information Model

Let

$$v_k(t) \in [0, 1] \quad (16)$$

denote the semantic importance of the information waiting for transmission to user k at slot t . The application layer may obtain $v_k(t)$ from an attention mechanism, object importance score, event priority, classifier confidence, task criticality, or another application-specific semantic metric. The transmitter supports a finite set of semantic representation levels

$$\mathcal{L} = \{1, \dots, L\}. \quad (17)$$

Let

$$y_{k\ell}(t) \in \{0, 1\} \quad (18)$$

indicate whether semantic representation ℓ is selected for user k . We impose

$$\sum_{\ell \in \mathcal{L}} y_{k\ell}(t) = 1, \quad \forall k, t. \quad (19)$$

Let $S_{k\ell}$ denote the number of information bits required by semantic representation ℓ . A more compact representation requires fewer radio resources but may reduce the achievable downstream task quality. We define

$$Q_{k\ell}(\gamma) \in [0, 1] \quad (20)$$

as the expected downstream task quality achieved when semantic representation ℓ is transmitted to user k under effective SINR γ . Depending on the application, $Q_{k\ell}$ may represent classification accuracy, detection probability, semantic similarity, reconstruction quality, or another normalized task-performance measure. The task utility delivered to user k becomes

$$U_k(t) = v_k(t) \sum_{\ell \in \mathcal{L}} y_{k\ell}(t) Q_{k\ell}(\gamma_k(t)). \quad (21)$$

Equation (21) creates an explicit connection between application semantics and physical-layer communication. Improving the reliability of a highly important semantic update produces greater system utility than achieving the same improvement for a low-value update.

1.6 MIMO Transmission and Resource Allocation

Let

$$a_{kn}(t) \in \{0, 1\} \quad (22)$$

indicate whether resource block n serves user k , and let $p_{kn}(t) \geq 0$ denote the corresponding transmit power. We use

$$q_{kb}(t) \in \{0, 1\} \quad (23)$$

to indicate the beam selected for user k . Each scheduled user selects one beam:

$$\sum_{b \in \mathcal{B}} q_{kb}(t) = 1. \quad (24)$$

The resulting beamforming vector is

$$\mathbf{v}_k(t) = \sum_{b \in \mathcal{B}} q_{kb}(t) \mathbf{w}_b. \quad (25)$$

For a multi-user MIMO transmission, the SINR experienced by user k on resource block n is

$$\gamma_{kn}(t) = \frac{a_{kn}(t)p_{kn}(t) |\mathbf{h}_k^H(t) \mathbf{v}_k(t)|^2}{\sigma_k^2 + \sum_{\substack{j \in \mathcal{K} \\ j \neq k}} a_{jn}(t)p_{jn}(t) |\mathbf{h}_k^H(t) \mathbf{v}_j(t)|^2}. \quad (26)$$

The achievable downlink rate of user k is therefore

$$R_k(t) = \sum_{n \in \mathcal{N}} W_n \log_2(1 + \gamma_{kn}(t)), \quad (27)$$

where W_n denotes the bandwidth of resource block n . If the gNB supports at most N_{RF} simultaneously multiplexed users on one resource block, we impose

$$\sum_{k \in \mathcal{K}} a_{kn}(t) \leq N_{\text{RF}}, \quad \forall n, t. \quad (28)$$

The power allocation must satisfy

$$0 \leq p_{kn}(t) \leq a_{kn}(t) P_{\max}, \quad (29)$$

and

$$\sum_{k \in \mathcal{K}} \sum_{n \in \mathcal{N}} p_{kn}(t) \leq P_{\max}, \quad \forall t. \quad (30)$$

The selected semantic payload must fit within the data transmitted during a scheduling interval of duration Δ :

$$\sum_{\ell \in \mathcal{L}} y_{k\ell}(t) S_{k\ell} \leq \Delta R_k(t), \quad \forall k, t. \quad (31)$$

1.7 Task-Value of CSI

An important question is not merely whether the gNB can improve its CSI estimate, but whether obtaining that improvement is worth the required feedback resources. Let $\hat{U}_k^0(t)$ denote the expected task utility if the gNB relies on beam prediction without receiving current CSI. Let $\hat{U}_k^m(t)$ denote the expected utility when user k provides feedback mode $m > 0$. We define the *Task-Value of CSI* (TVCSI) as

$$\text{TVCSI}_{k,m}(t) = \frac{\hat{U}_k^m(t) - \hat{U}_k^0(t)}{C_m + \epsilon}. \quad (32)$$

The TVCSI measures the expected increase in downstream semantic utility produced by one unit of CSI-feedback expenditure. Therefore, a user receives high-resolution CSI feedback only when the additional channel knowledge is expected to improve a valuable task sufficiently to justify its feedback cost. This metric creates the main cross-layer coupling in our framework:

$$\boxed{\text{Semantic Importance} \rightarrow \text{Value of Fresh CSI} \rightarrow \text{Feedback Resolution} \rightarrow \text{Beam Quality} \rightarrow \text{Task Performance.}} \quad (33)$$

1.8 Semantic Reliability

Average task utility alone may hide rare but severe failures. This issue becomes important when the system carries critical semantic information. Let $\mathcal{K}_c \subseteq \mathcal{K}$ denote users carrying critical information. For every such user, we impose

$$\Pr [Q_k(t) < Q_k^{\min}] \leq \varepsilon_k, \quad \forall k \in \mathcal{K}_c, \quad (34)$$

where

$$Q_k(t) = \sum_{\ell \in \mathcal{L}} y_{k\ell}(t) Q_{k\ell}(\gamma_k(t)). \quad (35)$$

Here, $Q_k^{\min}$ denotes the minimum acceptable semantic quality and ε_k denotes the maximum tolerated violation probability.

1.9 Risk-Aware Semantic Loss

We define the instantaneous importance-weighted semantic loss as

$$L^{\text{sem}}(t) = \sum_{k \in \mathcal{K}} v_k(t) [1 - Q_k(t)]. \quad (36)$$

To prevent the optimization from obtaining a high average utility while occasionally producing extremely poor semantic performance, we use the Conditional Value-at-Risk (CVaR) of $L^{\text{sem}}(t)$. For confidence level $\alpha \in (0, 1)$,

$$\text{CVaR}_\alpha(L^{\text{sem}}) = \min_{\eta} \left\{ \eta + \frac{1}{1 - \alpha} \mathbb{E} [(L^{\text{sem}} - \eta)^+] \right\}, \quad (37)$$

where $(x)^+ = \max\{x, 0\}$ and η represents the corresponding Value-at-Risk threshold.

1.10 Beam-Switching Cost

Frequent beam changes can increase beam-management overhead and make transmission less stable. We therefore define

$$s_k(t) = \mathbb{I}\{b_k(t) \neq b_k(t-1)\}, \quad (38)$$

where $\mathbb{I}\{\cdot\}$ denotes the indicator function.

1.11 Problem Statement

The gNB must decide how much channel information is worth acquiring before transmitting the semantic information currently waiting for each user. Requesting high-resolution CSI can improve beam selection and transmission reliability, but it consumes limited uplink feedback resources. Predicting the beam avoids this overhead but becomes unreliable under mobility or rapidly changing channels. At the same time, different semantic messages have different importance and require different levels of transmission fidelity. We therefore seek a policy that spends CSI-feedback resources only when the expected improvement in downstream task performance justifies the additional overhead. The policy must jointly select the CSI-feedback mode, semantic representation, transmission beam, resource blocks, and transmit power while satisfying radio-resource, feedback-capacity, and semantic-reliability constraints.

1.12 Joint Task-Value-Aware Optimization

We formulate the joint optimization problem as

$$\begin{aligned}
\mathbf{P1} : \quad & \max_{\mathbf{x}, \mathbf{y}, \mathbf{q}, \mathbf{a}, \mathbf{p}} \quad \liminf_{T \rightarrow \infty} \frac{1}{T} \sum_{t=1}^T \mathbb{E} \left[\sum_{k \in \mathcal{K}} U_k(t) \right. \\
& \quad - \lambda_F \sum_{k \in \mathcal{K}} \sum_{m \in \mathcal{M}} C_m x_{km}(t) \\
& \quad \left. - \lambda_B \sum_{k \in \mathcal{K}} s_k(t) \right] \\
& \quad - \lambda_R \text{CVaR}_\alpha(L^{\text{sem}}) \tag{39a} \\
\text{s.t.} \quad & \sum_{m \in \mathcal{M}} x_{km}(t) = 1, \quad \forall k, t, \tag{39b} \\
& \sum_{k \in \mathcal{K}} \sum_{m \in \mathcal{M}} C_m x_{km}(t) \leq C_{\max}^{\text{FB}}, \quad \forall t, \tag{39c} \\
& \sum_{\ell \in \mathcal{L}} y_{k\ell}(t) = 1, \quad \forall k, t, \tag{39d} \\
& \sum_{b \in \mathcal{B}} q_{kb}(t) = 1, \quad \forall k, t, \tag{39e} \\
& \sum_{k \in \mathcal{K}} a_{kn}(t) \leq N_{\text{RF}}, \quad \forall n, t, \tag{39f} \\
& 0 \leq p_{kn}(t) \leq a_{kn}(t) P_{\max}, \quad \forall k, n, t, \tag{39g} \\
& \sum_{k \in \mathcal{K}} \sum_{n \in \mathcal{N}} p_{kn}(t) \leq P_{\max}, \quad \forall t, \tag{39h} \\
& \sum_{\ell \in \mathcal{L}} y_{k\ell}(t) S_{k\ell} \leq \Delta R_k(t), \quad \forall k, t, \tag{39i} \\
& \Pr [Q_k(t) < Q_k^{\min}] \leq \varepsilon_k, \quad \forall k \in \mathcal{K}_c, \tag{39j} \\
& x_{km}(t), y_{k\ell}(t), q_{kb}(t), a_{kn}(t) \in \{0, 1\}, \quad \forall k, m, \ell, b, n, t, \tag{39k} \\
& p_{kn}(t) \geq 0, \quad \forall k, n, t. \tag{39l}
\end{aligned}$$

The first term in (39a) maximizes the importance-weighted downstream task quality. The second term penalizes CSI-feedback consumption and prevents the system from requesting high-resolution CSI from every user. The third term penalizes unnecessary beam switching, while the final CVaR term protects the system against the tail of severe semantic failures. Constraints (39b)–(39c) determine the CSI-feedback mode and enforce the shared feedback budget. Constraint (39d) selects one semantic representation for each user. Constraint (39e) selects the transmission beam. Constraints (39f)–(39h) enforce the spatial multiplexing and transmit-power limits. Constraint (39i) ensures that the selected semantic representation can be delivered within the scheduling interval. Constraint (39j) protects critical semantic traffic from unacceptable task-quality failures. Problem **P1** couples binary CSI-feedback decisions, semantic source selection, discrete beam selection, resource-block assignment, continuous power allocation, nonlinear MIMO rates, learned task-quality functions, and temporally correlated channel states. It is therefore a stochastic mixed-integer nonlinear optimization problem. Even simplified instances retain combinatorial feedback-allocation and resource-assignment subproblems. Solving **P1** exactly at every scheduling interval is consequently impractical for realistic network sizes. This structure motivates a hierarchical solution in which the system first learns task-oriented channel representations and beam predictions, then evaluates the value of obtaining additional CSI, and finally performs feedback scheduling and radio-resource allocation using lightweight optimization. Such a decomposition retains the interaction among semantic importance, CSI acquisition, beam management,

and resource allocation while keeping the online decision process suitable for practical 5G-Advanced operation.